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EXPERIMENT: VISCOSITY BY THE FALLING SPHERE METHOD

Objectives:

- 1. To determine the viscosity of liquids using the falling sphere method.

EXPERIMENT SUMMARY

INTRODUCTION

Every fluid has a characteristic thickness or viscosity, which is the measure of internal friction between fluid layers. The more viscous a fluid, the greater the drag force it exerts on an object moving through it. The device used to measure this viscosity value is called a viscometer [1].

In this experiment, viscosity is measured using the falling sphere method. When a small sphere is dropped into a fluid, it will initially experience acceleration due to gravity. However, the fluid's frictional or drag force will resist the sphere's motion until its acceleration decreases and eventually becomes zero. Under this zero-acceleration condition, the sphere travels at a constant velocity known as the terminal velocity (v_{term}) [1].

Physically, this terminal velocity is reached when the net force acting on the sphere is zero, or $\Sigma F = 0$. Three main forces act on the sphere as it falls within the fluid, namely the weight force (W) pulling the sphere downwards, as well as the Archimedes buoyant force (F_A) and the Stokes drag force (F_S) resisting the sphere's downward motion [1]. Additionally, the sphere's movement is assumed to be slow so that the fluid flow around the sphere is laminar [2].

A fluid's viscosity value is also highly influenced by temperature. In liquids, an increase in temperature will cause the distance between molecules to expand, thereby weakening the internal frictional force and decreasing the viscosity value [2]. Therefore, recording the room temperature is an important parameter during the experiment.

Based on the equilibrium of these forces, according to Stokes' law, the fluid viscosity (η) is formulated as [2]:

$$\eta = \frac{(\rho_{\text{sphere}} - \rho_{\text{fluid}})d_{\text{sphere}}^2 g}{18v_{\text{term}}} \quad (1)$$

where η is the fluid viscosity ($\text{Pa} \cdot \text{s}$), ρ_{sphere} is the density of the sphere (kg/m^3), ρ_{fluid} is the density of the fluid (kg/m^3), d_{sphere} is the diameter of the sphere (m), g is the gravitational acceleration (m/s^2), and v_{term} is the terminal velocity (m/s). The Stokes' law equation applies ideally if the fluid container's diameter is considered infinite or very large [2]. Considering that in practice the experiment is conducted inside a measuring cylinder, the tube walls will provide an additional drag effect that affects the falling speed of the sphere. Therefore, the equation needs to be adjusted with the Francis correction [3].

The Francis correction factor is influenced by the ratio between the sphere's diameter (d_{sphere}) and the tube's inner diameter (D_{tube}), which is formulated as [3]:

$$F = \left(1 - \frac{d_{\text{sphere}}}{D_{\text{tube}}}\right)^{2,25} \quad (2)$$

By incorporating the Francis correction factor into Stokes' law, the final equation for calculating viscosity becomes:

$$\eta_F = \eta_S F \quad (3)$$

where η_F is the viscosity resulting from the Francis correction and η_S is the viscosity from Stokes' law.

METHODOLOGY

A. Tools and Materials

The following are the tools and materials used in this experiment.

Table 1. Tools and materials for viscosity by the falling sphere method experiment

Tools and Materials	Quantity	Function
Falling sphere viscometer apparatus	1 set	As the main long tube-shaped container where the test fluid is filled and where the sphere falls to observe its movement.
Small spheres	3 pieces (different sizes)	As the test objects dropped into the fluid to measure their falling time.
Stopwatch	1 piece	To measure the sphere’s travel time while moving from the initial boundary point to the final boundary point.
Vernier caliper	1 piece	To measure the sphere’s diameter and the inside of the viscometer tube.
Latex/nitrile gloves	As needed	Personal protective equipment to prevent direct skin contact with the fluid and prevent dirt contamination on the equipment.
Digital balance	1 piece	To measure the mass of the sphere.
Retort stand	1 set	To support the alcohol thermometer.
Alcohol thermometer	1 piece	To measure the room temperature.
Cooking oil	As needed	As the first sample liquid whose viscosity value will be determined.
Engine oil	As needed	As the second sample liquid whose viscosity value will be determined.
Sphere retriever	1 set	To retrieve the sphere that has fallen to the bottom of the tube so it can be used for repetition.
Shop towels	As needed	To clean and dry fluid spills as well as dry the sphere before cleaning.
Cleaning solution	As needed	To clean the remnants of the test fluid adhering to the sphere.
Ballast	1 piece	To provide a load on the digital balance to make it more accurate.

B. Procedures

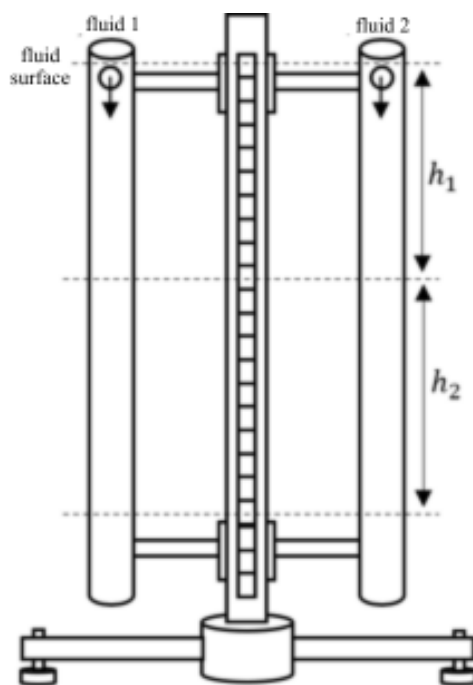


Figure 1. Schematic of the falling sphere viscometer apparatus

The experimental procedures are as follows:

1. Wear latex/nitrile gloves before starting the experiment.
2. Measure the room temperature using an alcohol thermometer.
3. Place the ballast in the middle of the digital balance weighing pan and record its mass (m_{ballast}). Place 5 plastic spheres of the same size inside the ballast and record the total mass (m_{total}). Calculate the mass of one plastic sphere using the formula $m_{\text{sphere}} = (m_{\text{total}} - m_{\text{ballast}})/5$. Perform mass measurements for all sizes.
4. Measure the diameter of the plastic sphere (d_{sphere}) for all sizes using the lower jaws of the vernier caliper.
5. Determine the initial point and final point of the sphere's fall time measurement on the tube. The height difference between these two points is calculated as the distance h . Ensure the initial point is far enough from the fluid surface so the sphere has reached terminal velocity when passing it.
6. Measure the inner tube diameter (D_{tube}) using the upper jaws of the vernier caliper.
7. Drop the sphere from the fluid surface and record the time required for the sphere to travel distance h using a stopwatch.
8. Retrieve the sphere by pulling the sphere retriever slowly from outside the tube.
9. Repeat steps 7 and 8 until 5 travel time data are obtained.
10. Repeat steps 7 to 9 for different sphere sizes in the same fluid.
11. Before switching to another test fluid, dry the spheres using shop towels then wash using the cleaning solution. Dry completely.
12. Repeat steps 6 to 11 for different fluids.

C. Experiment Flowchart

DATA ANALYSIS AND DISCUSSION

A. Data Analysis

The following are the tables of measurement data obtained in the laboratory:

Table 2. Measurement results for cooking oil fluid ($\rho_{\text{fluid}} = \dots\dots\dots \text{g/cm}^3$) with a height of $h = \dots\dots\dots \text{cm}$ and temperature $\dots\dots\dots ^\circ\text{C}$.

Sphere Variation	d_{sphere} (± mm)	m_{sphere} (± g)	t (± s)					D_{tube} (± mm)
			Repetition					
			1	2	3	4	5	
Small								
Medium								
Large								

Table 3. Measurement results for engine oil fluid ($\rho_{\text{fluid}} = \dots\dots\dots \text{g/cm}^3$) with a height of $h = \dots\dots\dots \text{cm}$ and temperature $\dots\dots\dots ^\circ\text{C}$.

Sphere Variation	d_{sphere} (± mm)	m_{sphere} (± g)	t (± s)					D_{tube} (± mm)
			Repetition					
			1	2	3	4	5	
Small								
Medium								
Large								

Approved by:

Laboratory Assistant

B. Calculation Analysis

Sample Calculation:

The following are the tables of calculated data obtained.

Table 4. Calculation results for cooking oil fluid ($\rho_{\text{fluid}} = \dots\dots\dots \text{g/cm}^3$) with a height of $h = \dots\dots\dots \text{cm}$ and temperature $\dots\dots\dots ^\circ\text{C}$.

Sphere Variation	ρ_{sphere} (g/cm ³)	v_{term} (cm/s)	η_s (mPa · s)				η_F (mPa · s)
			$\bar{\eta}_s$	σ_{η_s}	$\Delta\eta_s$	Result	
Kecil							
Sedang							
Besar							

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Table 5. Calculation results for engine oil fluid ($\rho_{\text{fluid}} = \dots\dots\dots \text{g/cm}^3$) with a height of $h = \dots\dots\dots \text{cm}$ and temperature $\dots\dots\dots \text{ }^\circ\text{C}$.

Sphere Variation	ρ_{sphere} (g/cm ³)	v_{term} (cm/s)	η_S (mPa · s)				η_F (mPa · s)
			$\bar{\eta}_S$	σ_{η_S}	$\Delta\eta_S$	Result	
Kecil							
Sedang							
Besar							

Note:

- $\bar{\eta}_S$: Mean value of η_S
- σ_{η_S} : Standard deviation of η_S
- $\Delta\eta_S$: Absolute uncertainty of η_S
- Report the result of η_S in the format: $\bar{\eta}_S \pm \Delta\eta_S$
- Report the result of η_F in the format: $\bar{\eta}_F \pm \Delta\eta_F$

C. Discussion

CONCLUSION

From this experiment, it can be concluded that:

- 1.

REFERENCES

- [1] D. Halliday, R. Resnick, and J. Walker, *Fundamentals of Physics*, 10th ed. Hoboken, NJ, USA: John Wiley & Sons, 2014.
- [2] B. R. Munson, D. F. Young, and T. H. Okiishi, *Fundamentals of Fluid Mechanics*, 6th ed. Hoboken, NJ, USA: John Wiley & Sons, 2009.
- [3] A. W. Francis, "Wall effect in falling ball method for viscosity," *Physics*, vol. 4, no. 11, pp. 403–406, 1933. doi: 10.1063/1.1745152.

PRELIMINARY ASSIGNMENTS

1. Draw the free-body diagram of a solid sphere moving downwards in a fluid. Based on Newton's second law, derive the equation for the sphere's velocity as a function of time $v(t)$ starting from when it is first dropped until it reaches terminal velocity.
2. Explain what is meant by the Reynolds number (Re) and what is the maximum allowable Re value range so that viscosity calculations using Stokes' law remain accurate?
3. Explain physically how the presence of the tube walls can affect the falling speed of the sphere.
4. Explain microscopically how an increase in temperature affects the viscosity value of a liquid fluid.